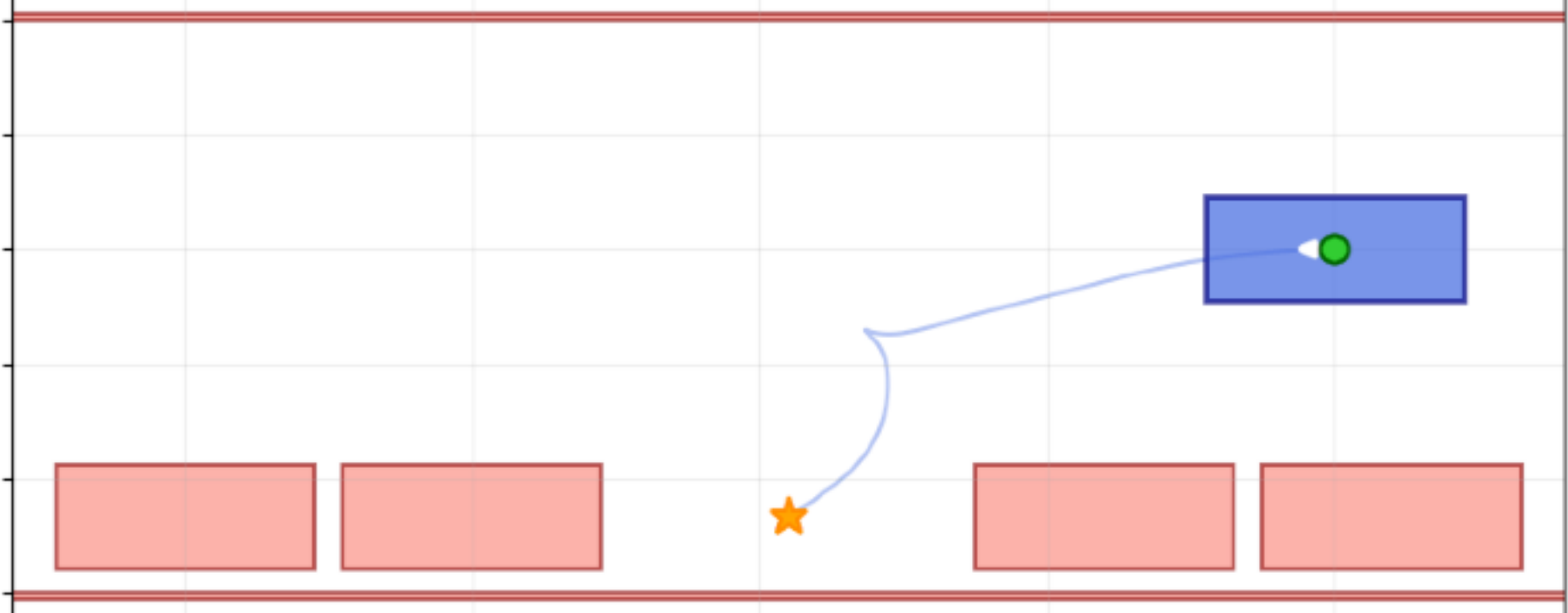


SE(2).NON-HOLONOMIC VEHICLES

A **left-invariant** $SE(2)$ metric encodes (soft) non-holonomic constraints directly into distance

— we plan in *continuous* $SE(2)$: no lattices or motion primitives are needed





Parallel Parking

Differential Drive

$$\xi = (v_x, v_y, \omega) \in \mathbb{R}^3 \cong \mathfrak{se}(2)$$

$$\langle \xi, \xi \rangle = \xi^\top G \xi, \quad G = \text{diag}(w_x, w_y, w_\theta)$$

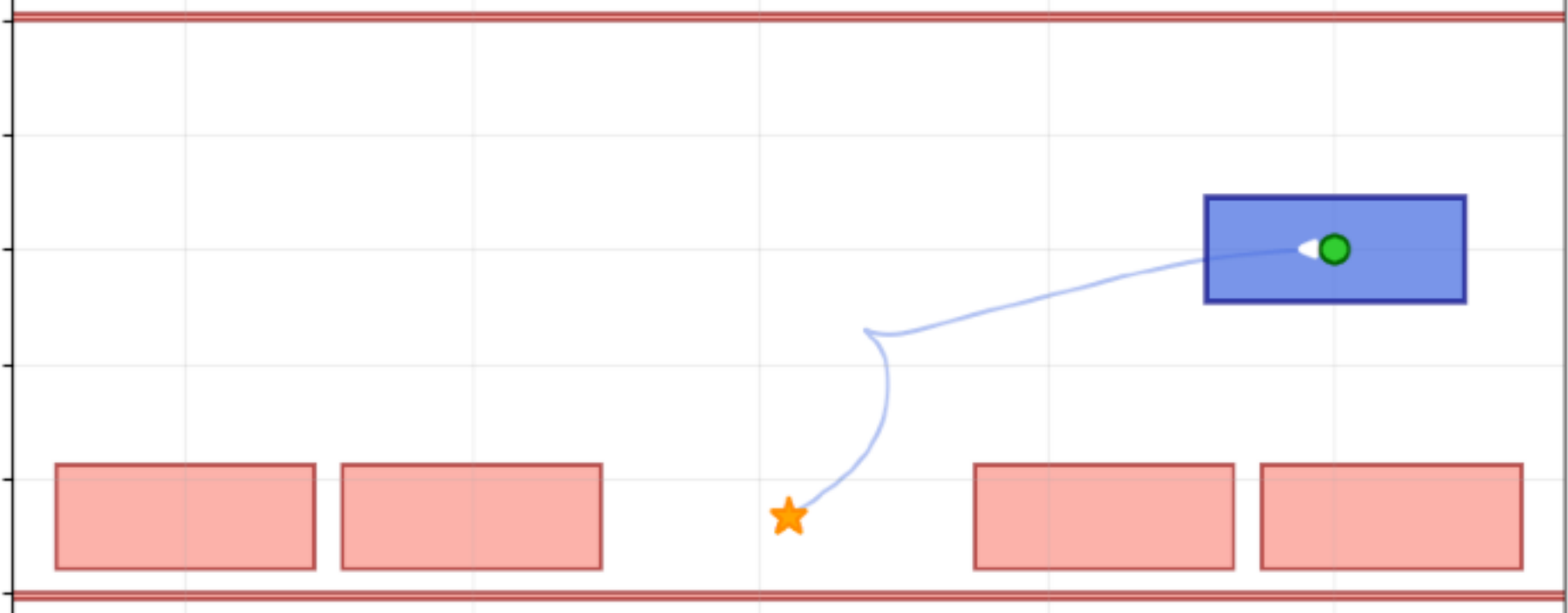


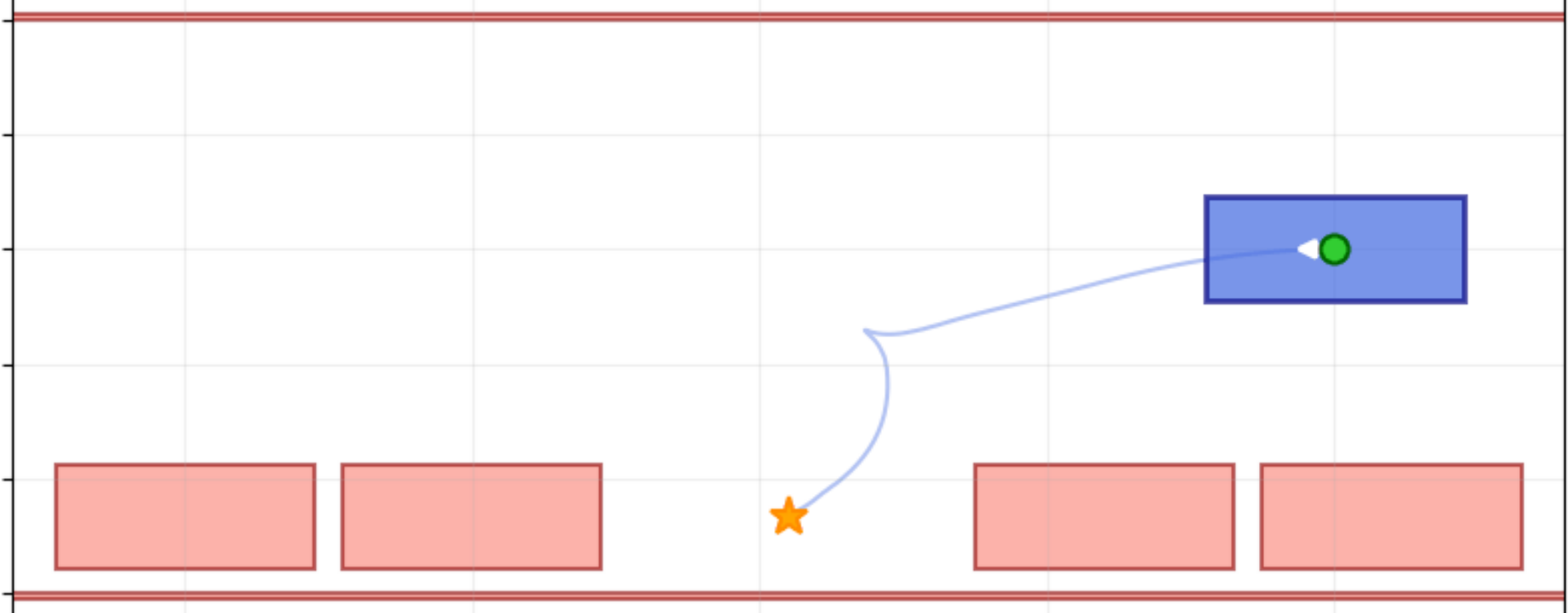
penalizes body-lateral motion

$$w_y \gg w_x$$

1

6







A **left-invariant** SE(2) metric encodes (soft) non-holonomic constraints directly into distance

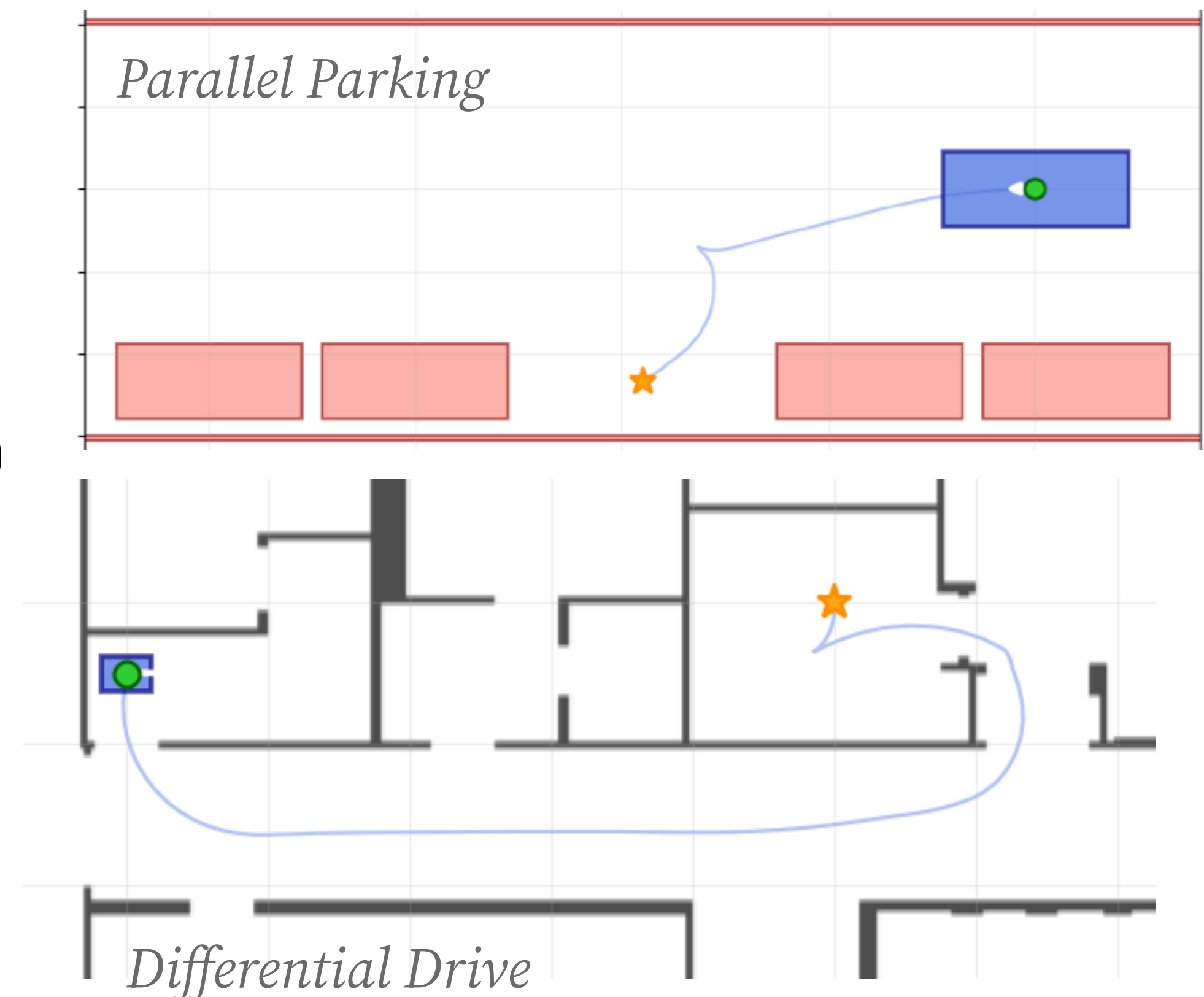
— we plan in *continuous* SE(2): no lattices or motion primitives are needed

$$\xi = (v_x, v_y, \omega) \in \mathbb{R}^3 \cong \mathfrak{se}(2)$$

$$\langle \xi, \xi \rangle = \xi^\top G \xi, \quad G = \text{diag}(w_x, w_y, w_\theta)$$

penalizes body-lateral motion

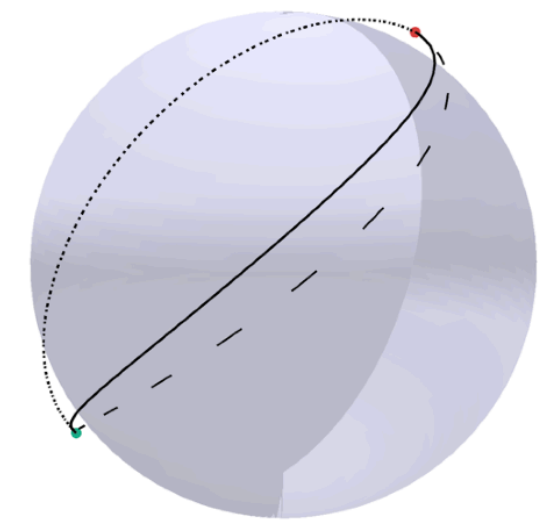
$$w_y \gg w_x$$



Where the method holds — and where it doesn't

1 The midpoint needs a **geodesically convex neighbourhood**

The two endpoints must be close enough that the log map is unique
(within the injectivity radius)



2 We plan on a Riemannian manifold (the **metric** $G(q)$ **is given**)

Not every non-Euclidean problem is Riemannian:

asymmetric costs $\rightarrow$ *Finsler*; hard non-holonomic constraints $\rightarrow$ *sub-Riemannian*